

Creation Date 06-Jul-2010

Revision Date 13-Oct-2023

Revision Number 4

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **SurePrep™ Plant/Fungi RNA Purification Kit, Lysis Solution**  
Cat No. : **BP2817-50B**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel,  
Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG,  
United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Acute oral toxicity  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation

Category 4 (H302)  
Category 2 (H315)  
Category 1 (H318)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### Hazard Statements

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H318 - Causes serious eye damage

### Precautionary Statements

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## 2.3. Other hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### 3.2. Mixtures

| Component | CAS No | EC No | Weight % | CLP Classification - Regulation (EC) No |
|-----------|--------|-------|----------|-----------------------------------------|
|-----------|--------|-------|----------|-----------------------------------------|

ACRBP2817B

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| 1272/2008                                     |           |                   |         |                                                                                                                                                                |
|-----------------------------------------------|-----------|-------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | 57-09-0   | EEC No. 200-311-3 | 2.5 - 5 | Acute Tox. 4 (H302)<br>Skin Irrit. 2 (H315)<br>Eye Dam. 1 (H318)<br>STOT SE 3 (H335)<br>STOT RE 2 (H373)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 1 (H410) |
| Trisodium citrate                             | 68-04-2   | 200-675-3         | 1 - 2   | -                                                                                                                                                              |
| Water                                         | 7732-18-5 | 231-791-2         | 10 - 25 | -                                                                                                                                                              |
| Guanidine, hydrochloride (1:1)                | 50-01-1   | EEC No. 200-002-3 | 65 - 75 | Acute Tox. 4 (H302)<br>Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)                                                                                             |

| Component                                     | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------------------------------------------|---------------------------------------|----------|-----------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | Skin Irrit. 2 : C ≥ 2.5 %             | 10       | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

Causes eye burns. Causes severe eye damage.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

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No information available.

## **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Hydrogen cyanide (hydrocyanic acid), Nitrogen oxides (NO<sub>x</sub>), Sulfur oxides.

## **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required.

### **6.2. Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### **7.2. Conditions for safe storage, including any incompatibilities**

Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 12

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### **7.3. Specific end use(s)**

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Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

#### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                                         | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-------------------------------------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide 57-09-0 ( 2.5 - 5 ) |                              |                                 |                                | DNEL = 0.4mg/kg bw/day            |
| Guanidine, hydrochloride (1:1) 50-01-1 ( 65 - 75 )                |                              |                                 |                                | DNEL = 1mg/kg bw/day              |

| Component                                                         | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-------------------------------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide 57-09-0 ( 2.5 - 5 ) | DNEL = 0.05mg/m <sup>3</sup>     |                                     |                                    |                                       |
| Guanidine, hydrochloride (1:1) 50-01-1 ( 65 - 75 )                |                                  | DNEL = 10.5mg/m <sup>3</sup>        |                                    | DNEL = 3.5mg/m <sup>3</sup>           |

#### Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                         | Fresh water      | Fresh water sediment | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|-------------------------------------------------------------------|------------------|----------------------|--------------------|------------------------------------|--------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide 57-09-0 ( 2.5 - 5 ) | PNEC = 0.022µg/L |                      | PNEC = 0.4µg/L     | PNEC = 0.19mg/L                    | PNEC = 0.21mg/kg soil dw |

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| Component                                                         | Marine water      | Marine water sediment | Marine water Intermittent | Food chain | Air |
|-------------------------------------------------------------------|-------------------|-----------------------|---------------------------|------------|-----|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide 57-09-0 ( 2.5 - 5 ) | PNEC = 0.0022µg/L |                       |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

#### Physical State

Liquid

#### Appearance

##### Odor

No information available

##### Odor Threshold

No data available

##### Melting Point/Range

No data available

##### Softening Point

No data available

##### Boiling Point/Range

No information available

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|                                                  |                          |                                   |
|--------------------------------------------------|--------------------------|-----------------------------------|
| Flammability (liquid)                            | No data available        |                                   |
| Flammability (solid,gas)                         | Not applicable           | Liquid                            |
| Explosion Limits                                 | No data available        |                                   |
| Flash Point                                      | No information available | Method - No information available |
| Autoignition Temperature                         | No data available        |                                   |
| Decomposition Temperature                        | No data available        |                                   |
| pH                                               | 6.5-8.5                  |                                   |
| Viscosity                                        | No data available        |                                   |
| Water Solubility                                 | No information available |                                   |
| Solubility in other solvents                     | No information available |                                   |
| Partition Coefficient (n-octanol/water)          |                          |                                   |
| Component                                        | log Pow                  |                                   |
| 1-Hexadecanaminium,<br>N,N,N-trimethyl-, bromide | 3.2                      |                                   |
| Guanidine, hydrochloride (1:1)                   | -1.7                     |                                   |
| Vapor Pressure                                   | No data available        |                                   |
| Density / Specific Gravity                       | No data available        |                                   |
| Bulk Density                                     | Not applicable           | Liquid                            |
| Vapor Density                                    | No data available        | (Air = 1.0)                       |
| Particle characteristics                         | Not applicable (liquid)  |                                   |

## 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization

Hazardous polymerization does not occur.

#### Hazardous Reactions

None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

Strong oxidizing agents. Acids.

### 10.6. Hazardous decomposition products

Hydrogen cyanide (hydrocyanic acid). Nitrogen oxides (NOx). Sulfur oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

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## Inhalation

Based on available data, the classification criteria are not met

## Toxicology data for the components

| Component                                     | LD50 Oral                | LD50 Dermal          | LC50 Inhalation   |
|-----------------------------------------------|--------------------------|----------------------|-------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | LD50 = 410 mg/kg ( Rat ) | -                    | -                 |
| Trisodium citrate                             | 5400 mg/kg (Mouse)       | -                    | -                 |
| Water                                         | -                        | -                    | -                 |
| Guanidine, hydrochloride (1:1)                | 475 mg/kg (Rat)          | >2000 mg/kg (Rabbit) | 3.2 mg/L/4h (Rat) |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs

No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

### Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Do not empty into drains. .

| Component                                     | Freshwater Fish               | Water Flea                   | Freshwater Algae                                            |
|-----------------------------------------------|-------------------------------|------------------------------|-------------------------------------------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide |                               |                              | EC50: = 0.09 mg/L, 96h<br>(Pseudokirchneriella subcapitata) |
| Trisodium citrate                             | LC50: 18000 - 32000 mg/L, 96h | EC50: 5600 - 10000 mg/L, 48h |                                                             |



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|  |                       |                 |  |
|--|-----------------------|-----------------|--|
|  | (Poecilia reticulata) | (Daphnia magna) |  |
|--|-----------------------|-----------------|--|

| Component                                     | Microtox                                          | M-Factor |
|-----------------------------------------------|---------------------------------------------------|----------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | = 9.84 mg/L EC50 Photobacterium phosphoreum 5 min | 10       |
| Trisodium citrate                             | EC50 1800 - 3200 mg/L 8 h                         |          |
| Guanidine, hydrochloride (1:1)                | = 88.7 mg/L EC50 Pseudomonas putida 18 h          |          |

**12.2. Persistence and degradability** No information available

**12.3. Bioaccumulative potential** No information available

| Component                                     | log Pow | Bioconcentration factor (BCF) |
|-----------------------------------------------|---------|-------------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | 3.2     | No data available             |
| Guanidine, hydrochloride (1:1)                | -1.7    | No data available             |

**12.4. Mobility in soil** No information available .

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

## 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                  |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer.                                                                                                                                                       |
| <b>Switzerland - Waste Ordinance</b>       | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

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**IMDG/IMO** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**ADR**

**14.1. UN number** UN1760  
**14.2. UN proper shipping name** Corrosive liquid, n.o.s  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** III

**IATA** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                     | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------------------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | 57-09-0   | 200-311-3 | -      | -   | X     | X    | KE-34534 | X    | X    |
| Trisodium citrate                             | 68-04-2   | 200-675-3 | -      | -   | X     | X    | KE-20843 | X    | X    |
| Water                                         | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |
| Guanidine, hydrochloride (1:1)                | 50-01-1   | 200-002-3 | -      | -   | X     | X    | KE-18111 | X    | X    |

| Component                                     | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------------------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | 57-09-0   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Trisodium citrate                             | 68-04-2   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water                                         | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Guanidine, hydrochloride (1:1)                | 50-01-1   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component | CAS No | REACH (1907/2006) - Annex XIV - Substances | REACH (1907/2006) - Annex XVII - Restrictions | REACH Regulation (EC 1907/2006) article 59 - |
|-----------|--------|--------------------------------------------|-----------------------------------------------|----------------------------------------------|
|-----------|--------|--------------------------------------------|-----------------------------------------------|----------------------------------------------|

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|                                               |           | Subject to Authorization | on Certain Dangerous Substances                                    | Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------------------|-----------|--------------------------|--------------------------------------------------------------------|----------------------------------------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | 57-09-0   | -                        | -                                                                  | -                                                        |
| Trisodium citrate                             | 68-04-2   | -                        | -                                                                  | -                                                        |
| Water                                         | 7732-18-5 | -                        | -                                                                  | -                                                        |
| Guanidine, hydrochloride (1:1)                | 50-01-1   | -                        | Use restricted. See item 75.<br>(see link for restriction details) | -                                                        |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                                     | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | 57-09-0   | Not applicable                                                                            | Not applicable                                                                           |
| Trisodium citrate                             | 68-04-2   | Not applicable                                                                            | Not applicable                                                                           |
| Water                                         | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           |
| Guanidine, hydrochloride (1:1)                | 50-01-1   | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 3 (self classification)

| Component                                     | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------------------------------------|---------------------------------------|-------------------------|
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide | WGK3                                  |                         |
| Trisodium citrate                             | WGK1                                  |                         |
| Guanidine, hydrochloride (1:1)                | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
|           |                                                                                                        |                                                                                 |                                                                                             |

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|                                                                      |                                      |  |  |
|----------------------------------------------------------------------|--------------------------------------|--|--|
|                                                                      | <b>814.81)</b>                       |  |  |
| 1-Hexadecanaminium, N,N,N-trimethyl-, bromide<br>57-09-0 ( 2.5 - 5 ) | Prohibited and Restricted Substances |  |  |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H400 - Very toxic to aquatic life

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 06-Jul-2010

**Revision Date** 13-Oct-2023

**Revision Summary** Not applicable.

# **SAFETY DATA SHEET**

SurePrep™ Plant/Fungi RNA Purification Kit, Lysis Solution

Revision Date 13-Oct-2023

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**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**